Programming Databases

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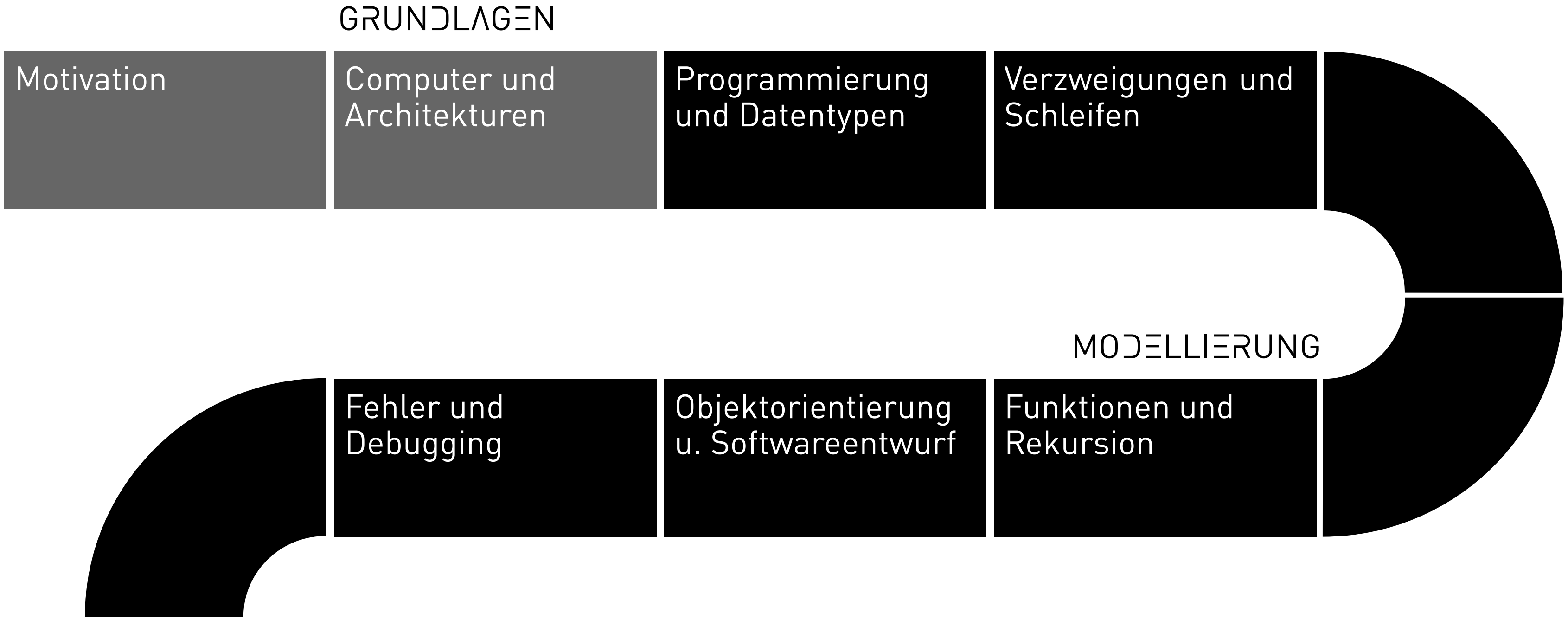
Programming Languages

MIDJOURNEY: P+D CODE RAIN, REF. MATRIX CODE RAIN

Workflow


GRUNDLAGEN
Motivation
Computer und Architekturen
Programmierung und Datentypen
Verzweigungen und Schleifen
MODELLIERUNG
Funktionen und Rekursion
Objektorientierung u. Softwareentwurf
Fehler und Debugging

Programming Languages

📘 **Definition: Programming Language**

A programming language is a formal language for creating programs for data processing on a computer. It is defined by its character set, syntax, and semantics.

The programming language enables humans (programmers) to create a program for the computer in a human-readable form.

Generations of Programming Languages

1. **Generation** - At first, computers were programmed directly in binary *machine code* that a processor could execute directly, but it could usually understand only a single processor type.

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1011 1111 1110 1000 1010
1101 0010 0000 0111 1111
```

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2. **Generation** – *assembly languages* use machine instructions (ADD, MOV, ...) for a specific processor type. The instructions are translated into machine code with the help of an assembler.

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MOV AL, 2        ; copies 2 into register AL
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3. **Generation** - *High-level programming languages* are human-readable and closer to natural language. A compiler translates the code into machine code.

```
while i < 20:
    x = x + i * i
    if x > 100:
        i = i + 3
```

The Most Popular Programming Languages

- New programming languages constantly evolve to meet changing requirements.
- Traditional application languages like C++ were replaced by server-side languages like Java.
- Later, there was a shift to web languages such as PHP, Perl, and JavaScript.
- In recent years, JavaScript has become popular for websites.
- Since 2017, Python has become increasingly popular due to its use in AI.

Python

- Python is the language of data science and AI
- Nowadays there are Python libraries for almost every application area
- Google Trends `data science` worldwide

Questions?